

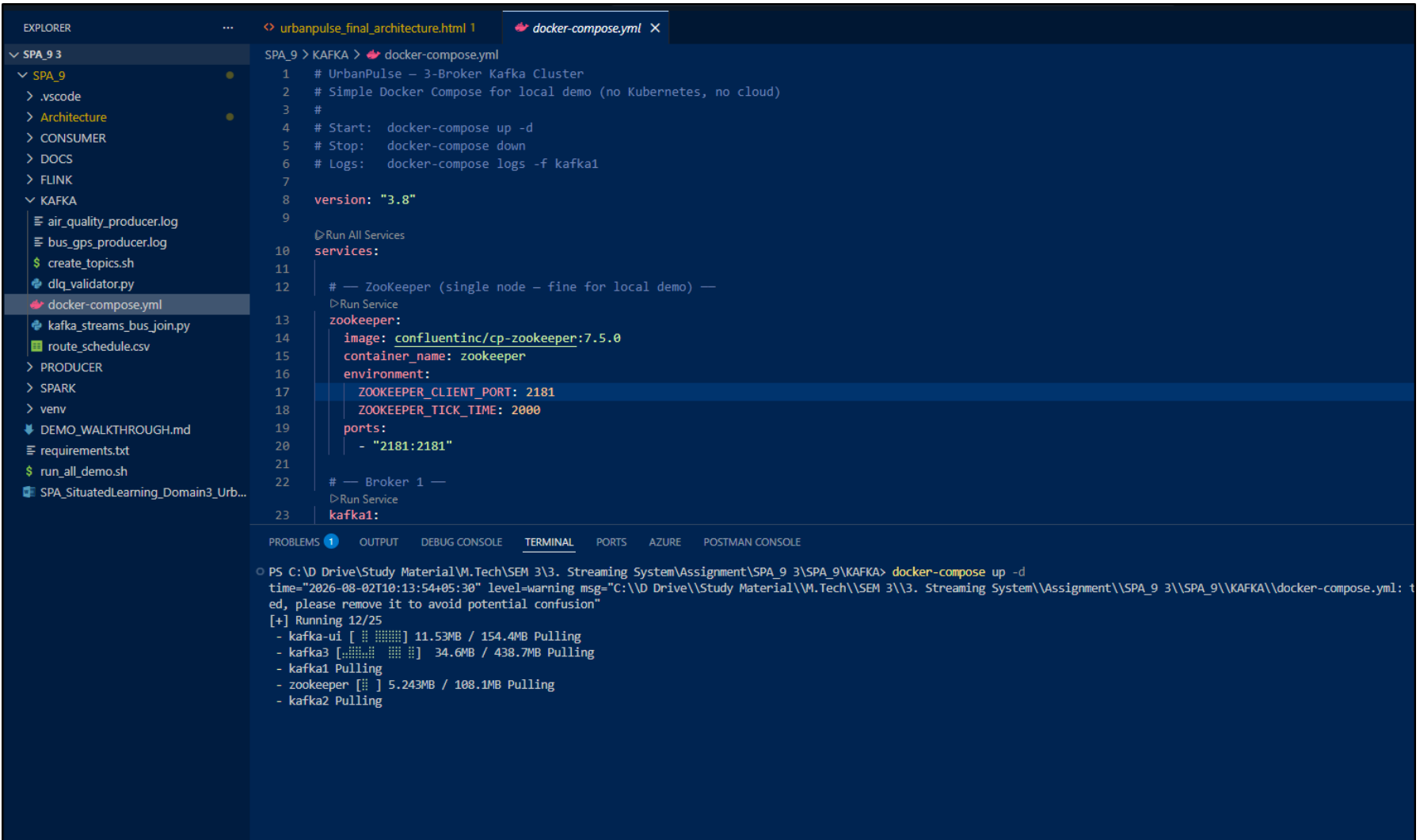
--TASK B: Apache Kafka — Multi-Source Urban Data Ingestion --

Set up a 3-broker Kafka cluster.

Step 1: Create Docker:

docker-compose.yml — Spins up a local 3-broker Kafka cluster (ZooKeeper-based, cp-kafka:7.5.0) with replication-factor 3 and min.insync.replicas 2 enforced cluster-wide; each broker exposes a dual-listener setup (INTERNAL for container-to-container traffic, EXTERNAL on ports 29092/29093/29094 for host access) and disables auto topic creation so topics are provisioned manually with justified partition/retention settings. Also includes Kafka UI (port 8080) for browsing topics/messages in the browser.

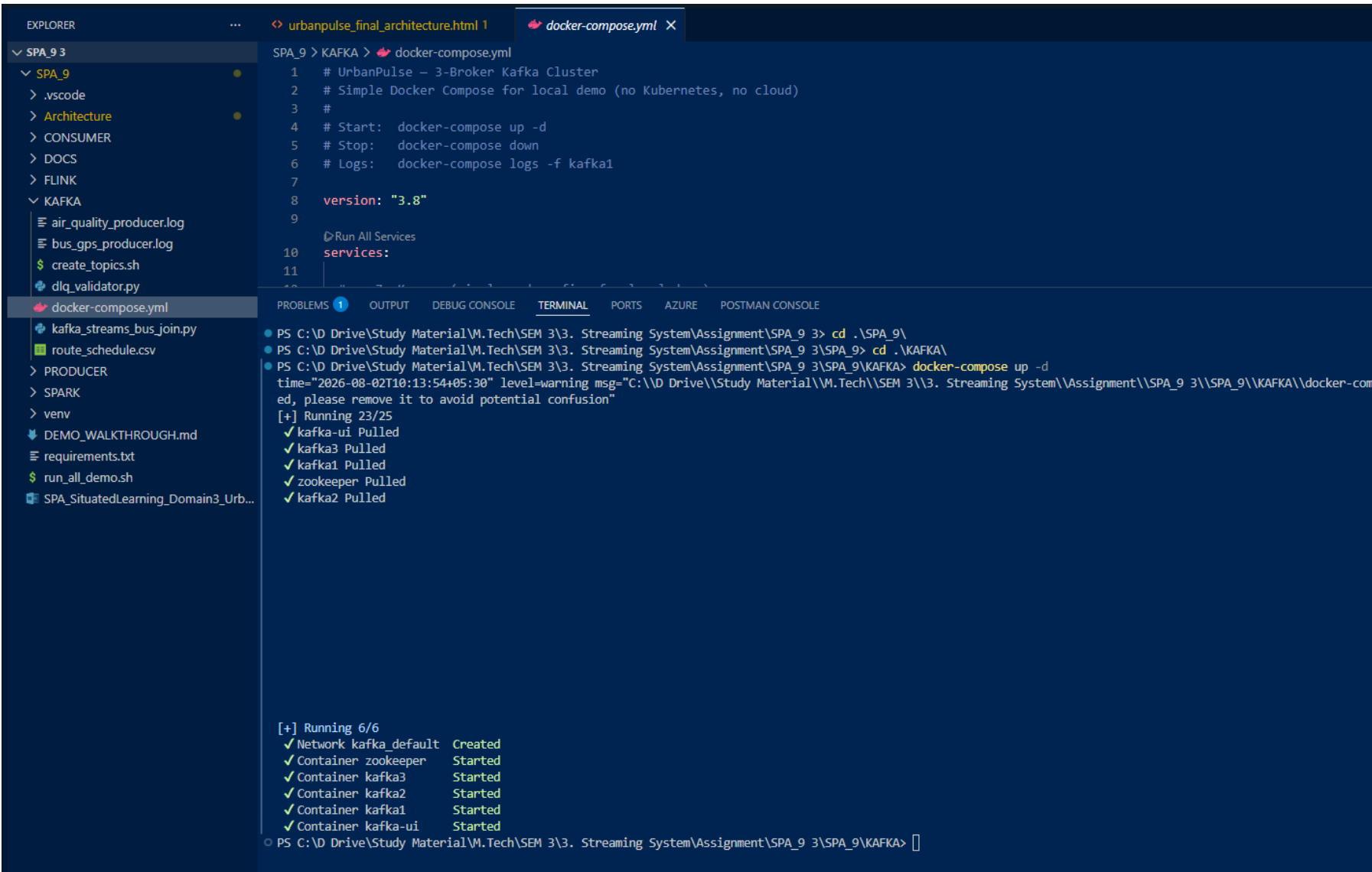
Command: docker-compose up -d



```
EXPLORER
  SPA_9_3
    SPA_9
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SPA_9 > KAFKA > docker-compose.yml
1 # UrbanPulse - 3-Broker Kafka Cluster
2 # Simple Docker Compose for local demo (no Kubernetes, no cloud)
3 #
4 # Start: docker-compose up -d
5 # Stop: docker-compose down
6 # Logs: docker-compose logs -f kafka1
7
8 version: "3.8"
9
10 Run All Services
11 services:
12   # -- ZooKeeper (single node - fine for local demo) --
13   Run Service
14   zookeeper:
15     image: confluentinc/cp-zookeeper:7.5.0
16     container_name: zookeeper
17     environment:
18       ZOOKEEPER_CLIENT_PORT: 2181
19       ZOOKEEPER_TICK_TIME: 2000
20     ports:
21       - "2181:2181"
22   # -- Broker 1 --
23   Run Service
24   kafka1:
```

```
PS C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9_3\SPA_9\KAFKA> docker-compose up -d
time="2026-08-02T10:13:54+05:30" level=warning msg="C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9_3\SPA_9\KAFKA\docker-compose.yml: t
ed, please remove it to avoid potential confusion"
[+] Running 12/25
- kafka-ui [███] 11.53MB / 154.4MB Pulling
- kafka3 [███] 34.6MB / 438.7MB Pulling
- kafka1 Pulling
- zookeeper [██] 5.243MB / 108.1MB Pulling
- kafka2 Pulling
```



```
EXPLORER
  SPA_9_3
    SPA_9
      .vscode
      Architecture
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        air_quality_producer.log
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      DEMO_WALKTHROUGH.md
      requirements.txt
      run_all_demo.sh
      SPA_SituatedLearning_Domain3_Urb...

SPA_9 > KAFKA > docker-compose.yml
1 # UrbanPulse - 3-Broker Kafka Cluster
2 # Simple Docker Compose for local demo (no Kubernetes, no cloud)
3 #
4 # Start: docker-compose up -d
5 # Stop: docker-compose down
6 # Logs: docker-compose logs -f kafka1
7
8 version: "3.8"
9
10 Run All Services
11 services:
12   # -- ZooKeeper (single node - fine for local demo) --
13   Run Service
14   zookeeper:
15     image: confluentinc/cp-zookeeper:7.5.0
16     container_name: zookeeper
17     environment:
18       ZOOKEEPER_CLIENT_PORT: 2181
19       ZOOKEEPER_TICK_TIME: 2000
20     ports:
21       - "2181:2181"
22   # -- Broker 1 --
23   Run Service
24   kafka1:
```

```
PS C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9_3\SPA_9\KAFKA> cd .\SPA_9\
PS C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9_3\SPA_9> cd .\KAFKA\
PS C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9_3\SPA_9\KAFKA> docker-compose up -d
time="2026-08-02T10:13:54+05:30" level=warning msg="C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9_3\SPA_9\KAFKA\docker-com
ed, please remove it to avoid potential confusion"
[+] Running 23/25
✓ kafka-ui Pulled
✓ kafka3 Pulled
✓ kafka1 Pulled
✓ zookeeper Pulled
✓ kafka2 Pulled

[+] Running 6/6
✓ Network kafka_default Created
✓ Container zookeeper Started
✓ Container kafka3 Started
✓ Container kafka2 Started
✓ Container kafka1 Started
✓ Container kafka-ui Started
PS C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9_3\SPA_9\KAFKA>
```

Kafka Cluster is running in the Docker Desktop: [Kafka brokers store data messages.]

- Kafka 1 (Broker 1)
- Kafka 2 (Broker 2)
- Kafka 3 (Broker 3)
- UI
- Zookeeper

docker.desktopPERSONAL

SearchCtrl+KSign in

ContainersGive feedback

Container CPU usage9.06% / 2200% (22 CPUs available)

Container memory usage1.92GB / 14.98GB

Show charts

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP ToolkitBETA

Docker Hub

Docker Scout

Extensions

Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU...	Memory usage...	Memory (%)	Disk read/write	Network I/O	PIDS	Actions
☐	wizardly_ganguly	764ca9e35797	ubuntu:latest		0%	0B / 0B	0%	0B / 0B	0B / 0B	0	
☐	Ubuntu	7ebb7280f4c1	ubuntu:latest		0%	0B / 0B	0%	0B / 0B	0B / 0B	0	
☐	MongoDB	c8bd38d91111	mongodb/mongodb-enterprise		0%	0B / 0B	0%	0B / 0B	0B / 0B	0	
☐	kind_bardeen	669920f72b17	mongodb/mongodb-enterprise		0%	0B / 0B	0%	0B / 0B	0B / 0B	0	
☐	n8n	82da06426db5	n8nio/n8n:latest	5888:5678	0%	0B / 0B	0%	0B / 0B	0B / 0B	0	
☒	kafka	-	-	-	9.06%	1.92GB / 76.7GB	12.52%	365.66MB / 12.81	4.09MB / 8.72MB	86, 96, 92, 9	
☐	ui	5851b95b20e2	provectuslabs/kafka-ui:latest	8080:8080	0.18%	469.8MB / 15.34%	2.99%	101MB / 1.11MB	1.48MB / 5.04MB	86	
☐	kafka3	577c09532630	confluentinc/cp-kafka:7.5.0	29094:29094	2.83%	454.8MB / 15.34%	2.9%	5.37MB / 3.08MB	820KB / 1.3MB	96	
☐	kafka2	72d171608212	confluentinc/cp-kafka:7.5.0	29093:29093	2.55%	431.7MB / 15.34%	2.75%	705KB / 3.03MB	751KB / 1.03MB	92	
☐	kafka1	16dc4881508a	confluentinc/cp-kafka:7.5.0	29092:29092	3.22%	465.1MB / 15.34%	2.96%	190MB / 3.66MB	777KB / 1.06MB	92	
☐	zookeeper	39bb0abc2d8f	confluentinc/cp-zookeeper:7.5	2181:2181	0.28%	144.9MB / 15.34%	0.92%	68.6MB / 1.93MB	328KB / 301KB	104	

Step 2: We can see the APACHE KAFKA UI

UI for Apache Kafka

localhost:8080

MicrosoftLenovoSocialAIM.TechDSASOracleAWSOthersStudyAIKanbanFlowGoogleLog in | MongoDBIOB - Internet Banki...Royal Enfield Classic...Excalidraw Whitebo...

UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Dashboard

Online1 clustersOffline0 clusters

Only offline clusters

Cluster name	Version	Brokers count	Partitions	Topics	Production
urbanpulse-local	3.5-IV2	3	0	0	0 Bytes

UI for Apache Kafka

localhost:8080/ui/clusters/urbanpulse-local/brokers

MicrosoftLenovoSocialAIM.TechDSASOracleAWSOthersStudyAIKanbanFlowGoogleLog in | MongoDBIOB - Internet Banki...Royal Enfield Classic...Excalidraw Whitebo... (160) 1- Introductio...

UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Brokers

Uptime

Partitions

Broker Count3Active Controller3Version3.5-IV2

Online0 of 0URP0In Sync Replicas0 of 0Out Of Sync Replicas0

Broker ID	Disk usage	Partitions skew	Leaders	Leader skew	Online partitions	Port	Host
1	N/A	-	-	-	-	9092	kafka1
2	N/A	-	-	-	-	9092	kafka2
3	N/A	-	-	-	-	9092	kafka3

Step 3: Create Topic:

- **create_topics.ps1** — Creates all 9 UrbanPulse Kafka topics (bus GPS, traffic signals, air quality, smart meters, incidents, DLQ, health advisories, ward energy summary, and enriched bus GPS) with replication-factor 3, topic-specific partition counts, and justified retention periods (ranging from 24 hours to 365 days based on regulatory/operational needs).
- Runs via docker exec against the kafka1 container using the internal broker address, then lists all created topics for verification.

Run Command: create_topics.ps1

Topic	Purpose
urbanpulse.bus_gps	Raw GPS pings from all city buses, keyed by route, retained 24h for same-day ETA replay and accident reconstruction.
urbanpulse.traffic_signals	Live traffic signal states from all junctions, keyed by junction, retained 7 days for weekly signal-timing optimization.
urbanpulse.air_quality	Pollution sensor readings across city zones, retained 90 days for CPCB quarterly AQI trend reporting.
urbanpulse.smart_meters	Household/ward electricity meter readings, retained 365 days to meet Electricity Act 2003 audit and DISCOM submission requirements.
urbanpulse.incidents	Real-time alerts emitted by Flink (e.g. gridlock, anomalies), retained 30 days for incident history.
urbanpulse.dlq	Dead-letter queue capturing malformed/unprocessable messages from any pipeline stage, retained 7 days for ops debugging.
urbanpulse.health_advisories	Public health advisories generated by Spark SQL from air-quality trends, retained 30 days.
urbanpulse.ward_energy_summary	Aggregated per-ward energy consumption summaries from Spark streaming, retained 30 days for dashboard consumption.
urbanpulse.bus_gps_enriched	Bus GPS data enriched (e.g. joined with route/schedule info) via Kafka Streams, retained 24h for real-time ETA feeds.

EXPLORER

urbanpulse_final_architecture.html

create_topics.sh

create_topics.ps1 1 X

SPA_9 3

SPA_9

.vscode

Architecture

CONSUMER

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air_quality_producer.log

bus_gps_producer.log

create_topics.ps1 1

create_topics.sh

dlq_validator.py

docker-compose.yml

kafka_streams_bus_join.py

route_schedule.csv

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requirements.txt

run_all_demo.sh

SPA_SituatedLearning_Domain3_Urb...

SPA_9 > KAFKA > create_topics.ps1 > ...

100

101 # -----

102 # Topic 8: urbanpulse.ward_energy_summary (Spark streaming output)

103 # Partitions: 10 - one per 6 wards (60 wards total)

104 # Retention: 30 days

105 # -----

106 Create-Topic -Name "urbanpulse.ward_energy_summary" -Partitions 10 -RetentionMs 2592000000 -Reason "30 days - dashboard feed"

107

108 # -----

109 # Topic 9: urbanpulse.bus_gps_enriched (Kafka Streams Join output)

110 # Partitions: 40 - one per route (matching input)

111 # Retention: 24 hours

112 # -----

113 Create-Topic -Name "urbanpulse.bus_gps_enriched" -Partitions 40 -RetentionMs 86400000 -Reason "24 h - enriched ETA feed"

114

115 Write-Host ""

116 Write-Host "=====

117 Write-Host " All topics created. Listing:"

118 Write-Host "=====

119 docker exec kafka1 kafka-topics --bootstrap-server \$Broker --list

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE POSTMAN CONSOLE

PS C:\D Drive\Study Material\W.Tech\SEM 3\3. Streaming System\Assignment\SPA_9 3\SPA_9\KAFKA> .\create_topics.ps1

=====

UrbanPulse - Creating Kafka Topics

=====

> Creating: urbanpulse.bus_gps

Partitions : 40

Retention : 86400000 ms (24 h - accident replay window)

WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.

Created topic urbanpulse.bus_gps.

> Creating: urbanpulse.traffic_signals

Partitions : 15

Retention : 604800000 ms (7 days - weekly signal optimisation)

WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.

Created topic urbanpulse.traffic_signals.

> Creating: urbanpulse.air_quality

Partitions : 10

Retention : 777600000 ms (90 days - quarterly pollution trend)

WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.

Created topic urbanpulse.air_quality.

> Creating: urbanpulse.smart_meters

Partitions : 20

Retention : 3153600000 ms (365 days - regulatory energy audit)

WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.

Created topic urbanpulse.smart_meters.

EXPLORER

urbanpulse_final_architecture.html

create_topics.ps1 1

dlq_validator.py 1

SPA_9 3

SPA_9

create_topics.ps1

create_topics.sh

dlq_validator.py

docker-compose.yml

kafka_streams_bus_join.py

route_schedule.csv

PRODUCER

air_quality_producer.py

bus_gps_producer.py

smart_meter_producer.py

traffic_signals_producer.py

SPARK

venv

DEMO_WALKTHROUGH.md

requirements.txt

run_all_demo.sh

SPA_SituatedLearning_Domain3_Urb...

106

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119

Create-Topic -Name "urbanpulse.ward_energy_summary" -Partitions 10 -RetentionMs 259200000 -Reason "30 days - dashboard feed"

Topic 9: urbanpulse.bus_gps_enriched (Kafka Streams Join output)

Partitions: 40 - one per route (matching input)

Retention: 24 hours

Create-Topic -Name "urbanpulse.bus_gps_enriched" -Partitions 40 -RetentionMs 86400000 -Reason "24 h - enriched ETA feed"

Write-Host ""

Write-Host "=====

Write-Host " All topics created. Listing:"

Write-Host "=====

docker exec kafka1 kafka-topics --bootstrap-server \$Broker --list

PROBLEMS 2

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

AZURE

POSTMAN CONSOLE

[2026-08-02 05:18:56,822] WARN [AdminClient clientId=adminclient-1] Connection to node 3 (localhost/127.0.0.1:29094) could not be established. Broker not available

Error while executing topic command : Timed out waiting for a node assignment. Call: createTopics

[2026-08-02 05:18:57,264] ERROR org.apache.kafka.common.errors.TimeoutException: Timed out waiting for a node assignment. Call: createTopics (kafka.admin.TopicCommand\$)

=====

All topics created. Listing:

=====

urbanpulse.air_quality

urbanpulse.bus_gps

urbanpulse.bus_gps_enriched

urbanpulse.dlq

urbanpulse.health_advisories

urbanpulse.incidents

urbanpulse.smart_meters

urbanpulse.traffic_signals

urbanpulse.ward_energy_summary

PS C:\D Drive\Study Material\M.Tech\SEM 3\3. Streaming System\Assignment\SPA_9 3\SPA_9\KAFKA>

We can see 9 topics go created on the Dashboard:

localhost:8080

UI for Apache Kafka 83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Dashboard

Online 1 clusters

Offline 0 clusters

Only offline clusters

Cluster name	Version	Brokers count	Partitions	Topics	Production	Consumption
urbanpulse-local	3.5-IV2	3	148	9	0 Bytes	0 Bytes

localhost:8080/ui/clusters/urbanpulse-local/all-topics?perPage=25

UI for Apache Kafka 83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Topics

Search by Topic Name

Show Internal Topics

Delete selected topics

Copy selected topic

Purge messages of selected topics

Topic Name	Partitions	Out of sync replicas	Replication Factor	Number of messages	Size
urbanpulse.air_quality	10	0	3	0	0 Bytes
urbanpulse.bus_gps	40	0	3	0	0 Bytes
urbanpulse.bus_gps_enriched	40	0	3	0	0 Bytes
urbanpulse.dlq	3	0	3	0	0 Bytes
urbanpulse.health_advisories	5	0	3	0	0 Bytes
urbanpulse.incidents	5	0	3	0	0 Bytes
urbanpulse.smart_meters	20	0	3	0	0 Bytes
urbanpulse.traffic_signals	15	0	3	0	0 Bytes
urbanpulse.ward_energy_summary	10	0	3	0	0 Bytes

← → ↺

localhost:8080/ui/clusters/urbanpulse-local/brokers

🔍 ☆ 🗑️ |

🗄️ | 📁 Microsoft 📁 Lenovo 📁 Social 📁 AI 📁 M.Tech 📁 DSA 📁 Oracle 📁 AWS 📁 Others 📁 StudyAI 📁 KanbanFlow 📁 Google 🟢 Log in | MongoDB 🔵 IOB - Internet Banki... 🚗 Royal Enfield Classic... 📄 Excalidraw Whitebo... 📺 (160) 1- Introductio...

🔵 UI for Apache Kafka 83b5a60 v0.7.2

🔔

Dashboard

urbanpulse-lo... ^

Brokers

Topics

Consumers

Brokers

Uptime

Broker Count
3

Active Controller
3

Version
3.5-IV2

Partitions

Online
148 of 148

URP
0

In Sync Replicas
444 of 444

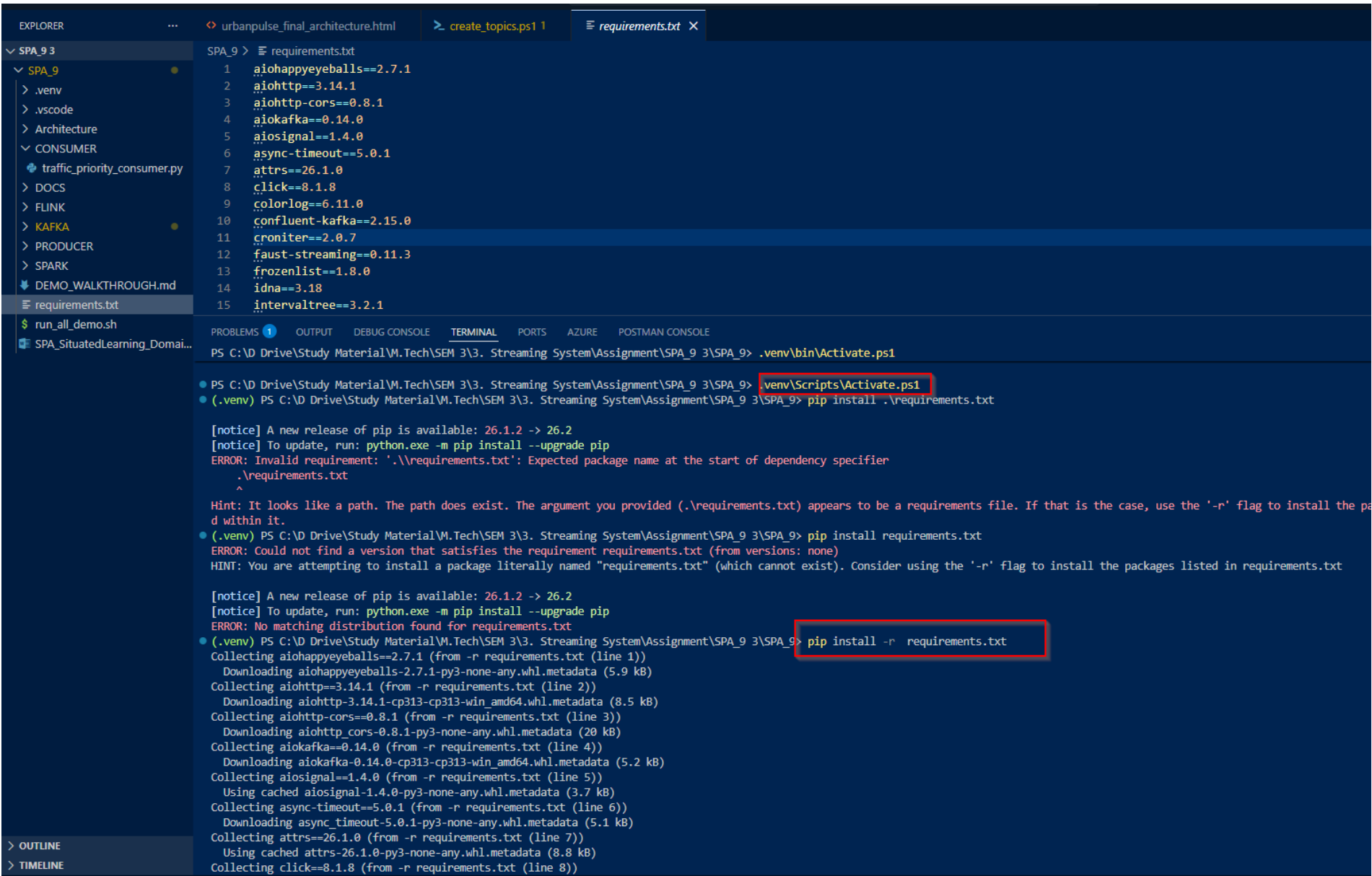
Out Of Sync Replicas
0

↕ Broker ID	↕ Disk usage	↕ Partitions skew ⓘ	↕ Leaders	↕ Leader skew	↕ Online partitions	↕ Port	↕ Host
1	N/A	-	47	-4.70%	148	9092	kafka1
2	N/A	-	50	1.40%	148	9092	kafka2
3 🟢	N/A	-	51	3.40%	148	9092	kafka3

Implement Python producers for the bus_gps and air_quality streams:

Step 4: Create Virtual environment and install requirement.txt:

- Command: python -m venv .venv (create virtual environment)
- Command: pip install -r requirement.txt (enable required libraries)
- Command: .venv\Scripts\Activate.ps1 (Activate the virtual environment)

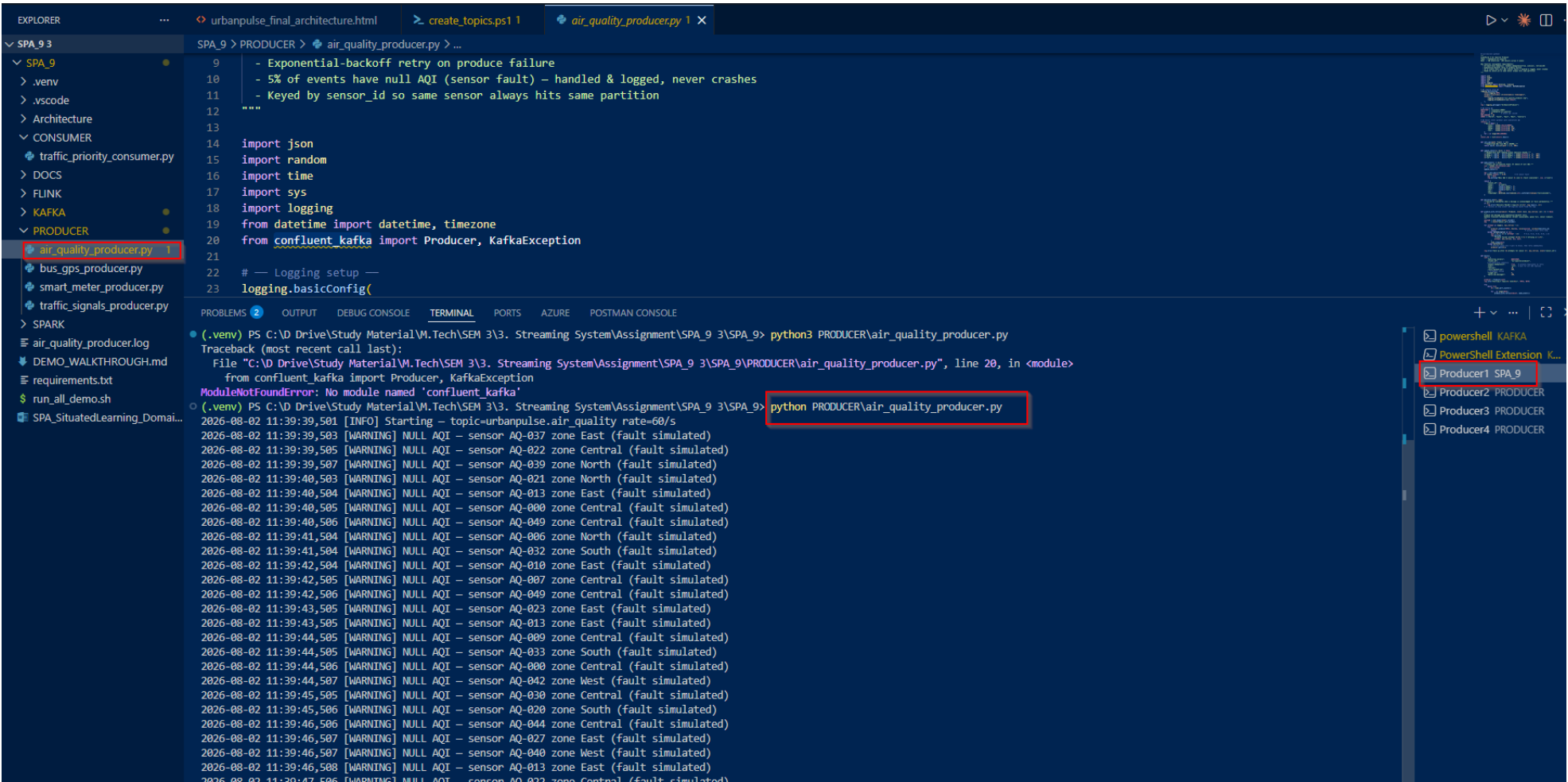


Script explanation:

- **bus_gps_producer.py** — Simulates 600 buses across 40 routes emitting GPS pings to urbanpulse.bus_gps at ~2,400 events/sec, keyed by route_id for per-route ordering, with idempotent at-least-once delivery (acks=all, retries, exponential backoff).
- **smart_meter_producer.py** — Simulates 800 smart meters across 60 wards emitting readings (kWh, voltage, power factor) to urbanpulse.smart_meters at ~1,100 events/sec, keyed by meter_id, with idempotent at-least-once delivery.
- **traffic_signals_producer.py** — Simulates 120 traffic junctions across 5 zones emitting signal state (vehicle count, wait time, phase) to urbanpulse.traffic_signals at ~380 events/sec, keyed by junction_id, with idempotent at-least-once delivery.
- **air_quality_producer.py** — Simulates 50 sensors across 5 zones emitting pollution readings to urbanpulse.air_quality at ~60 events/sec, keyed by sensor_id; deliberately injects 5% null-AQI events to simulate sensor faults for downstream DLQ validation testing, with idempotent at-least-once delivery.

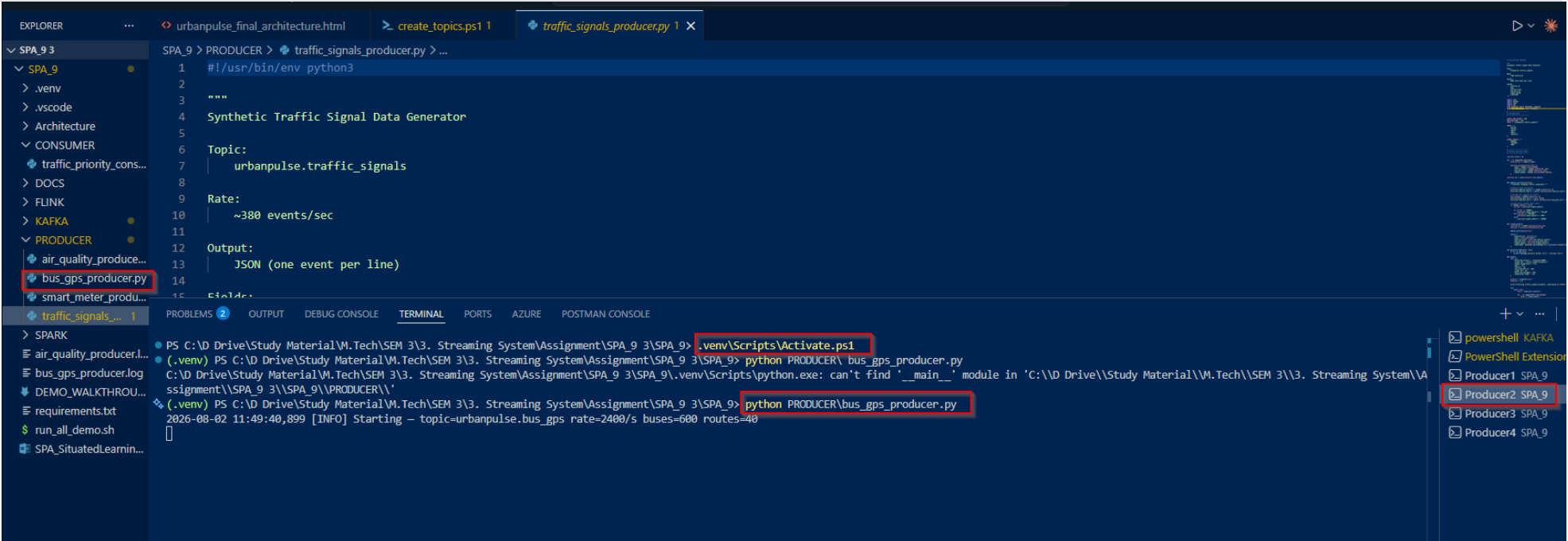
- Producer script for [smart_meter_producer](#)

Command python PRODUCER\air_quality_producer.py



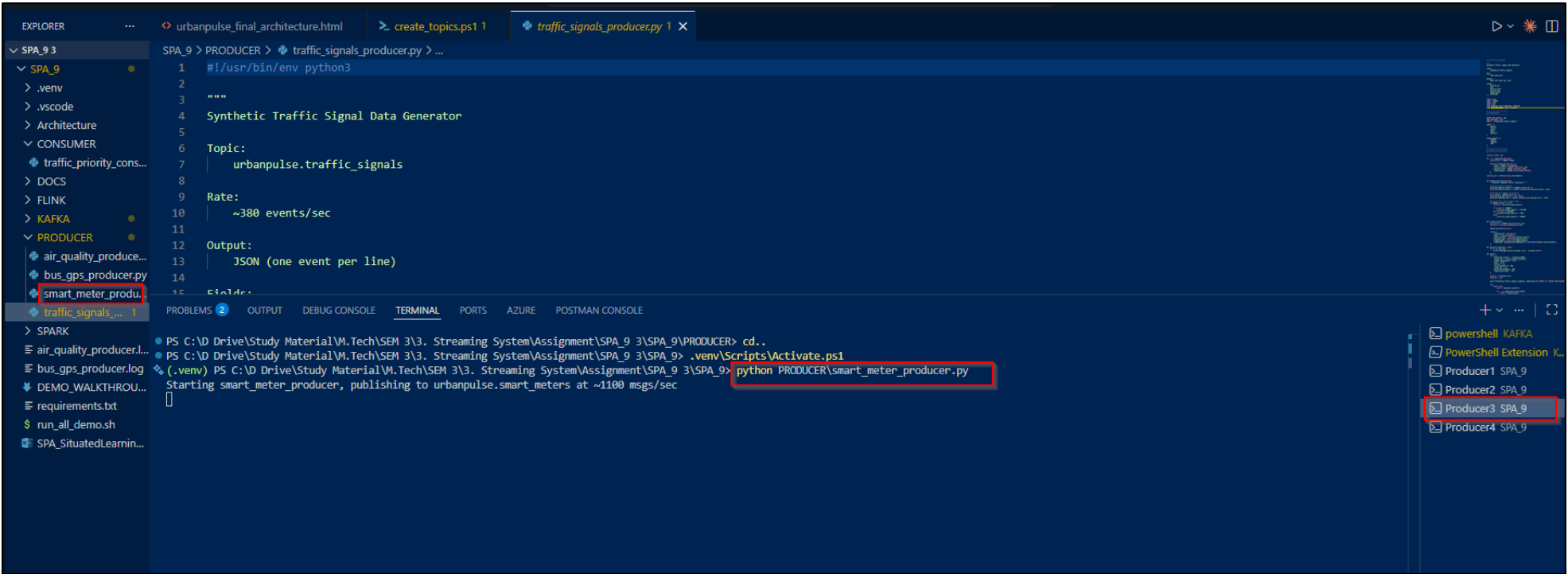
- Producer script for [smart_meter_producer](#)

Command: python PRODUCER\bus_gps_producer.py



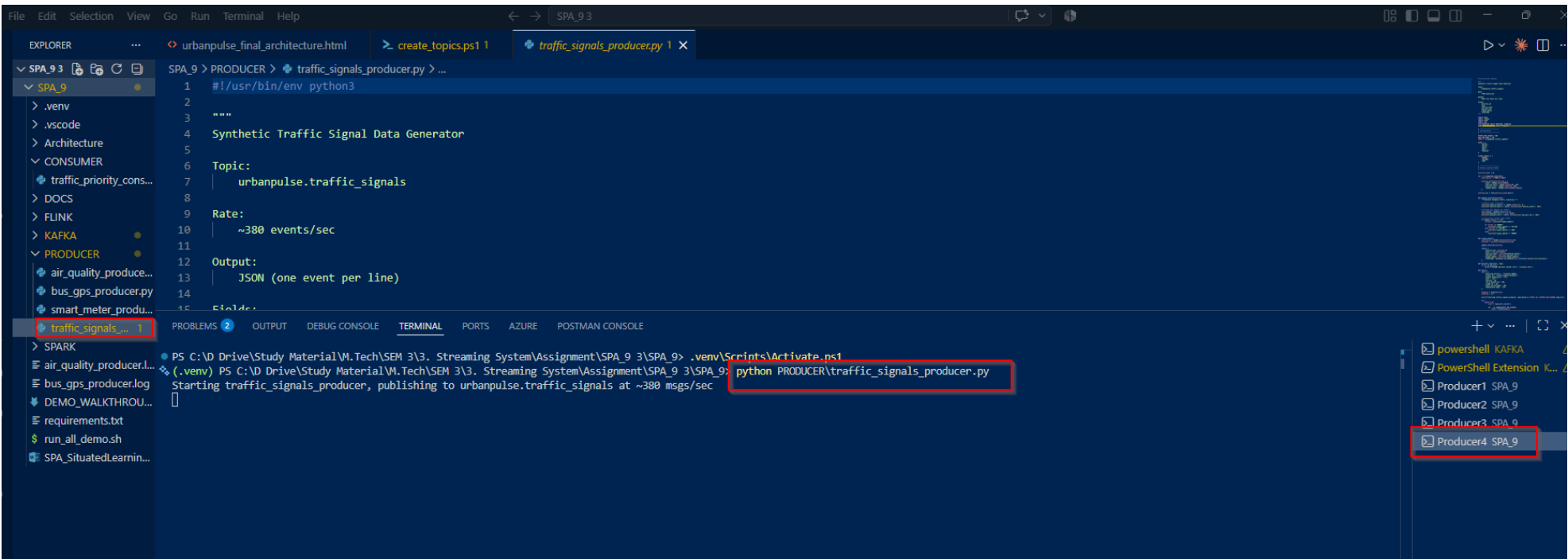
- Producer script for [smart_meter_producer](#)

Command: python PRODUCER\smart_meter_producer.py

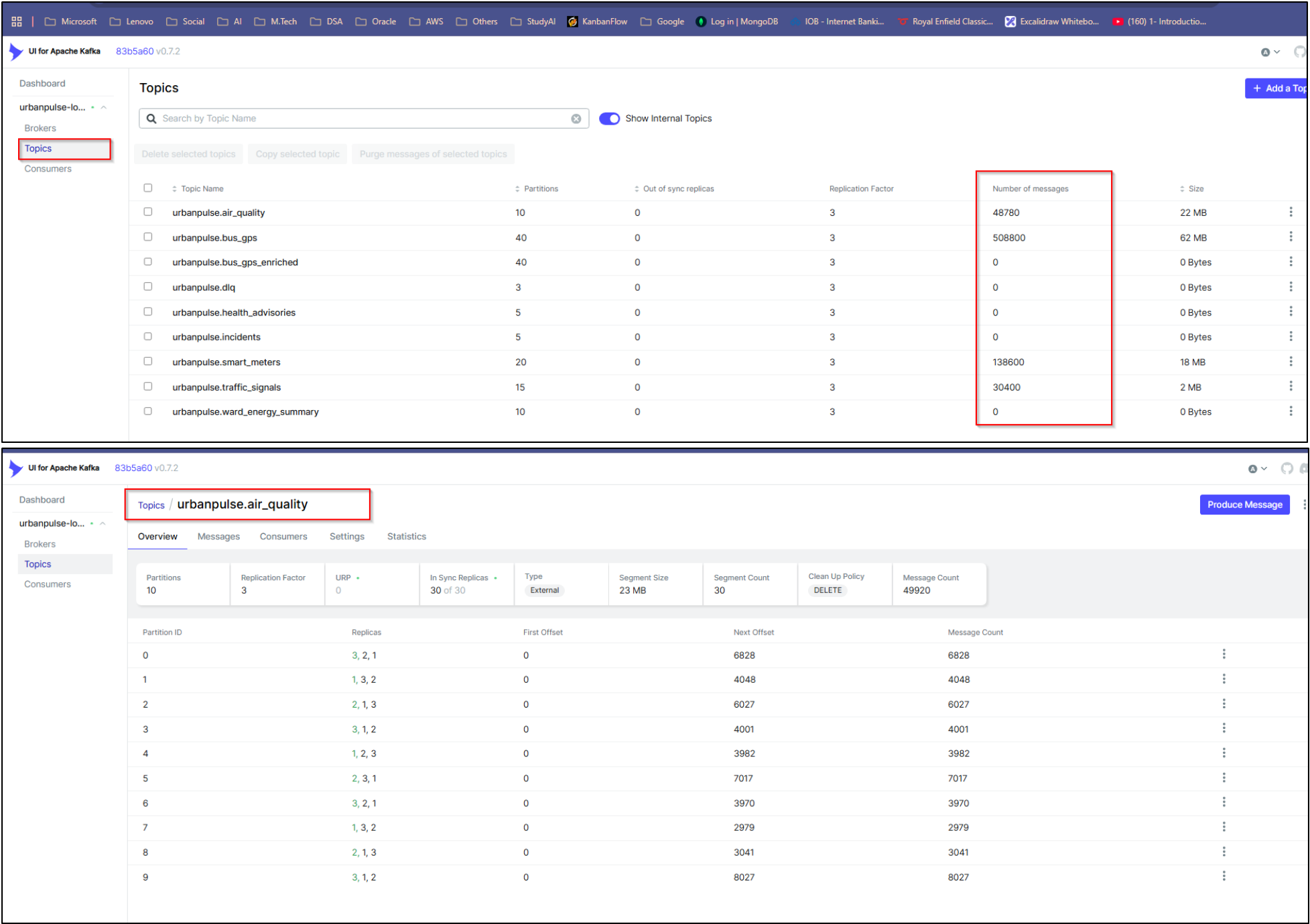


- Producer script for [Traffic Signals](#)

Command: python PRODUCER\traffic_signals_producer.py



Step 5: We can see the ‘[ProducerScript](#)’ are writing the message in the ‘[topics](#)’:

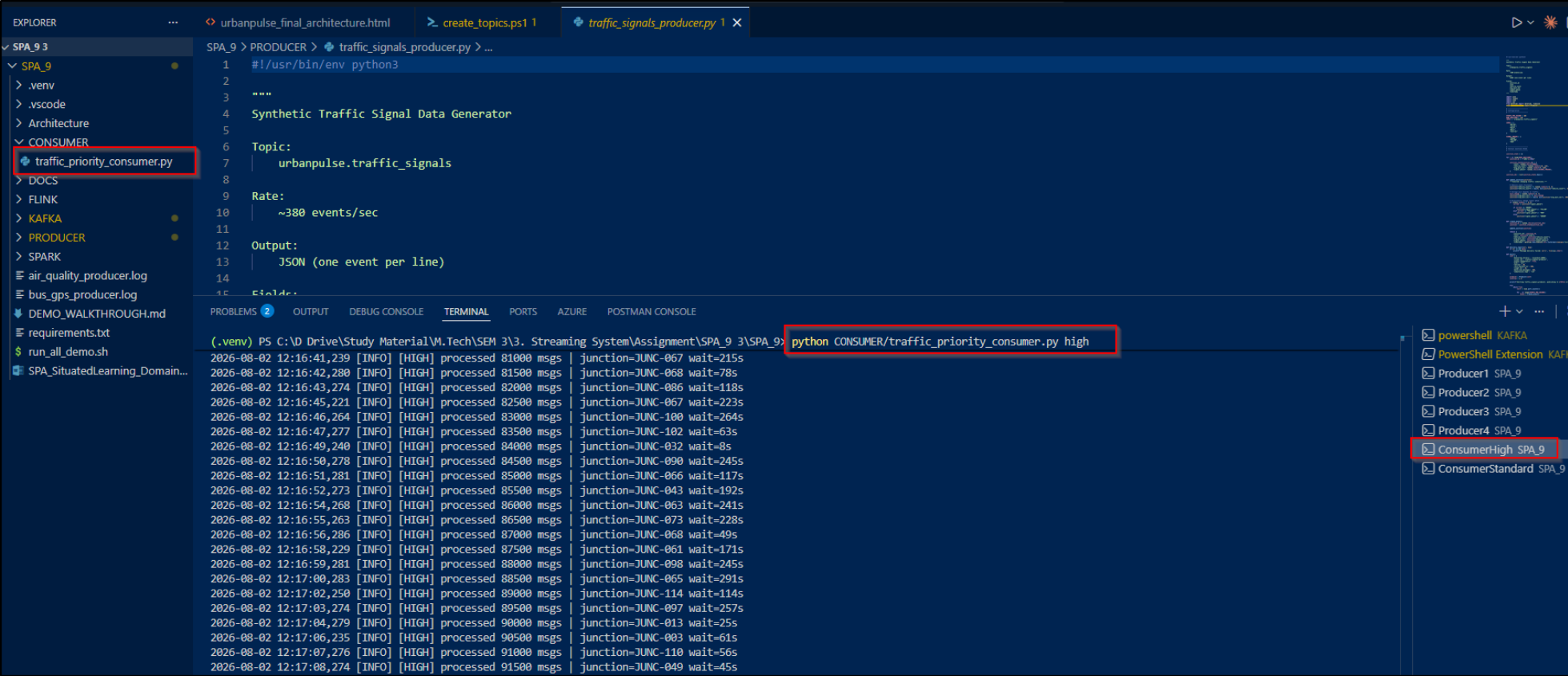


Build a priority consumer architecture:

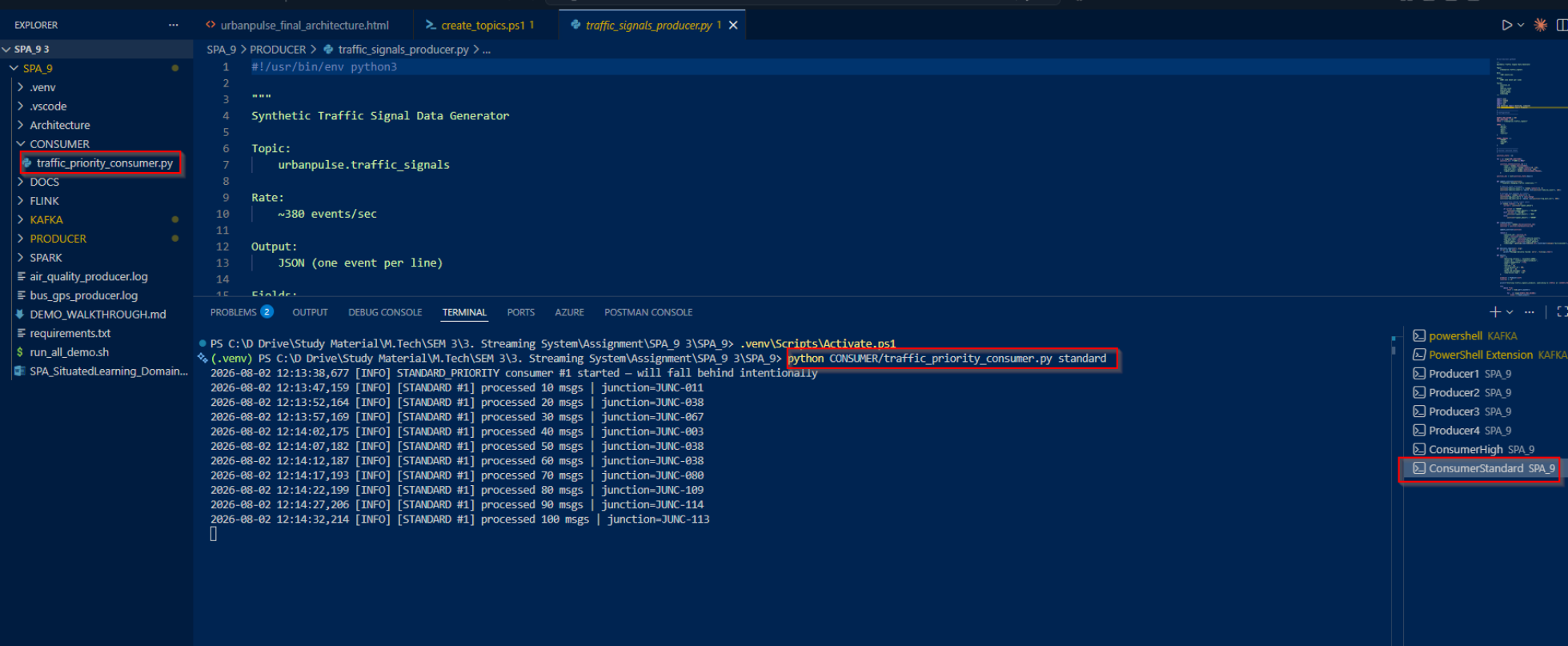
- **traffic_priority_consumer.py** — Demonstrates consumer-group isolation on `urbanpulse.traffic_signals` using two independent groups: **HIGH_PRIORITY_GROUP** (single fast consumer simulating real-time signal control, stays near-zero lag) and **STANDARD_PRIORITY_GROUP** (multiple consumers with an artificial 0.5s delay simulating slower analytics processing, deliberately falls behind).
- Since Kafka tracks offsets per consumer group independently, **it proves that a slow/lagging group has zero impact** on a separate group reading the same topic.

Step 6: Run the python script to create priority consumer architecture:

Command: python CONSUMER/traffic_priority_consumer.py high



Command: python CONSUMER/traffic_priority_consumer.py high



We can see the ‘lag’ in consumer log between ‘High_Priority_Group’ and ‘Standard_Priority_Group’:

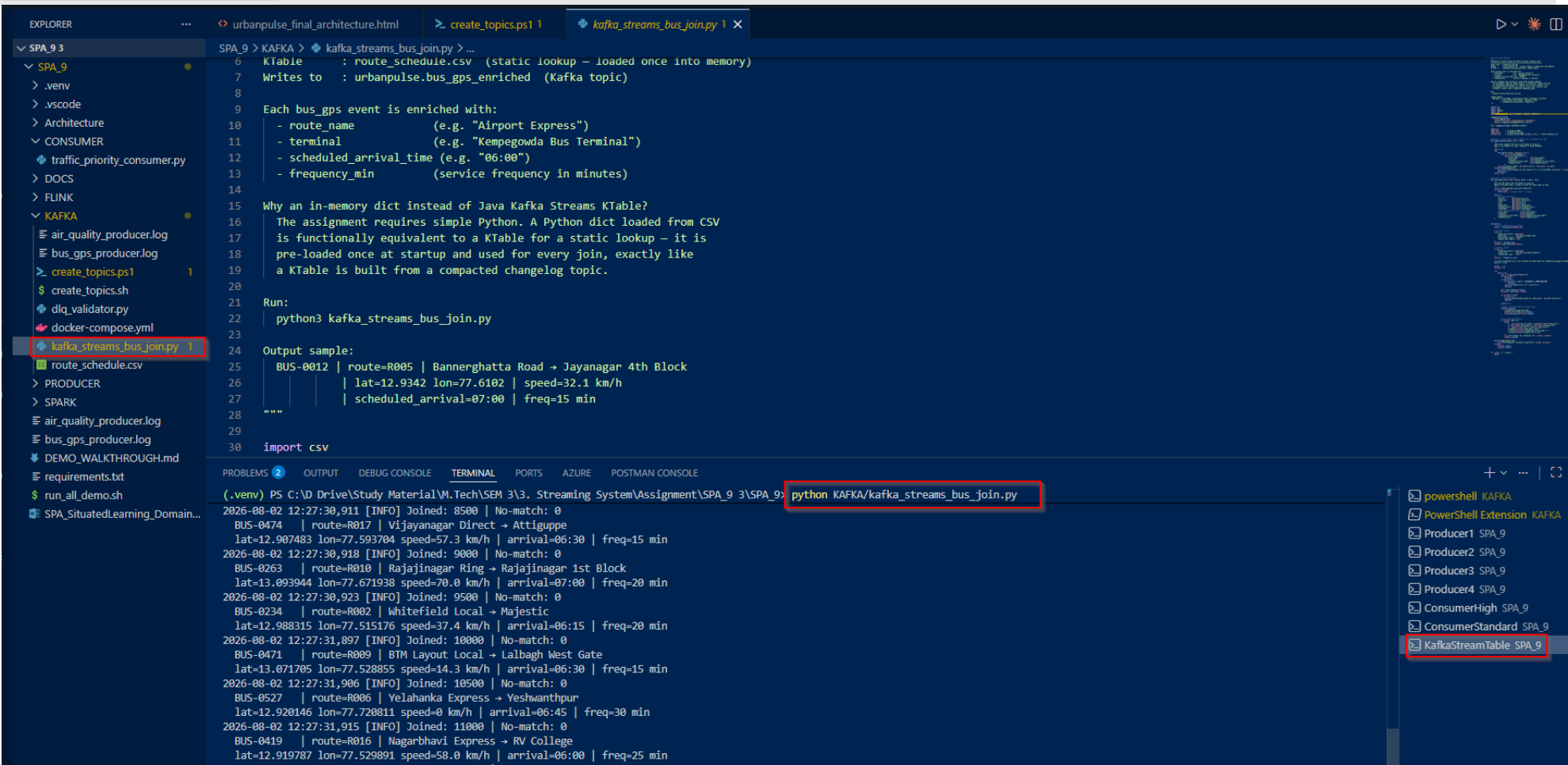
UI for Apache Kafka 83b5a60 v0.7.2					
Dashboard					
urbanpulse-lo...					
Brokers					
Topics					
Consumers					
Consumers					
Search by Consumer Group ID					
Group ID		Num Of Members	Num Of Topics	Consumer Lag	Coordinator
HIGH_PRIORITY_GROUP		1	1	1520	3
STANDARD_PRIORITY_GROUP		1	1	8684	1
					State
					STABLE
					STABLE

Build a Kafka Streams application that performs real-time enrichment

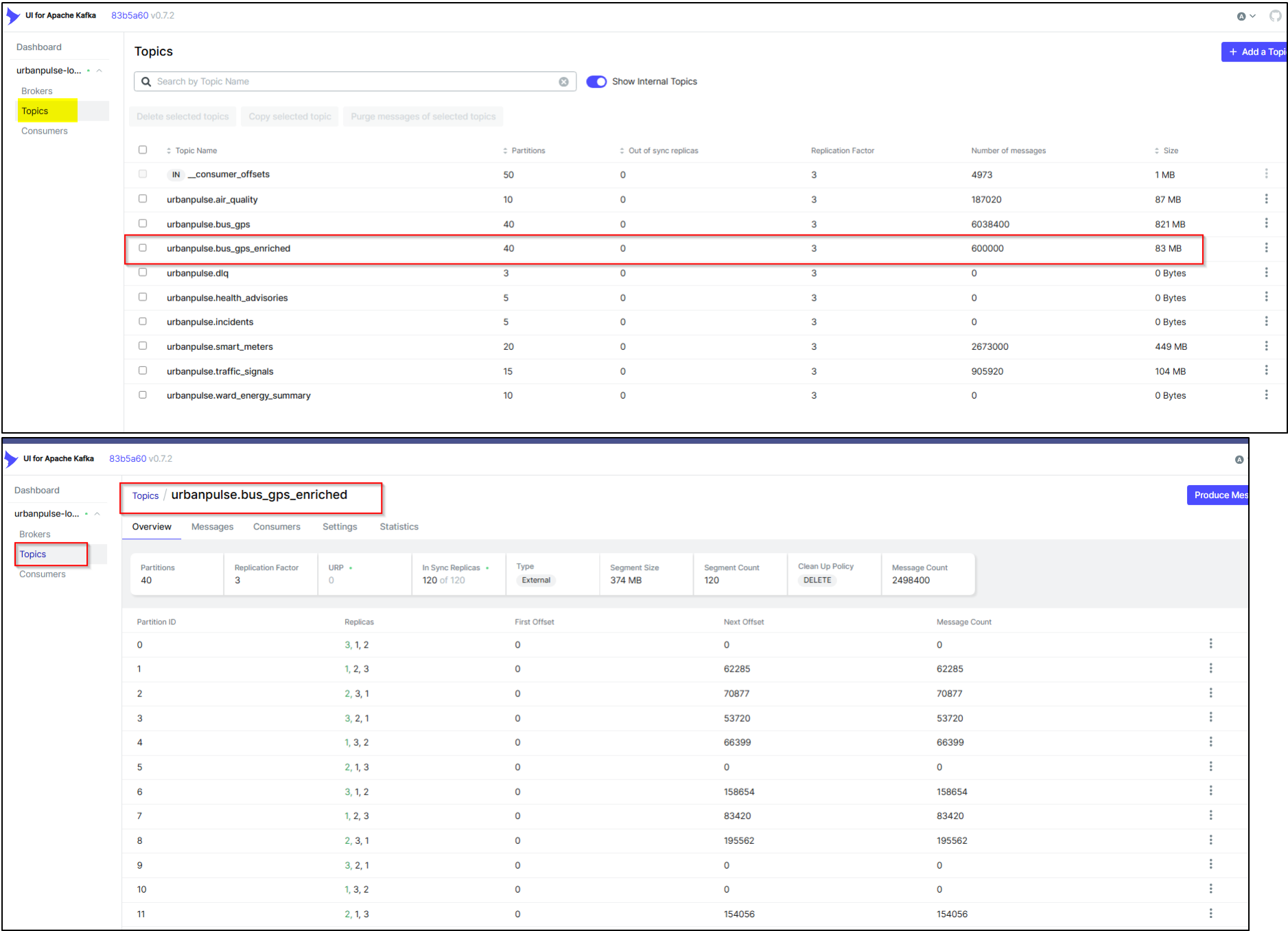
- This **Kafka Streams join enriches each event** with the human-readable route name, terminal, and scheduled arrival time from our static KTable — which is loaded once from route_schedule.csv at startup. In production, a KTable in Java Kafka Streams is built from a compacted changelog topic.
- Our Python equivalent is a dict — functionally identical for static lookups. The enriched events go to `urbanpulse.bus_gps_enriched`, which powers the passenger-facing real-time bus app

Step 7: Run the python script to enrich real-time Kafka Streams data:

Command: python KAFKA/ kafka_streams_bus_join.py



We can see the enrich data in Topics:



UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Topicsurbanpulse.bus_gps_enriched

OverviewMessagesConsumersSettingsStatistics

Seek Type

Offset

Offset

Partitions

All items are selected.

Key Serde

String

Value Serde

String

Clear all

Submit

Q Search

+ Add Filters

DONE

204 ms

	Offset	Partition	Timestamp	KeyPreview	ValuePreview
+	0	18	8/2/2026, 12:27:27	R037	{ "bus_id": "BUS-0220", "route_id": "R037", "lat": 1...
+	1	18	8/2/2026, 12:27:27	R037	{ "bus_id": "BUS-0273", "route_id": "R037", "lat": 1...
-	2	18	8/2/2026, 12:27:27	R037	{ "bus_id": "BUS-0585", "route_id": "R037", "lat": 1...

KeyValueHeaders

```
{  "bus_id": "BUS-0585",  "route_id": "R037",  "lat": 13.065889,  "lon": 77.734029,  "speed_kmh": 70.9,  "occupancy_pct": 35,  "timestamp": "2026-08-02T06:57:27.832+00:00",  "route_name": "Panathur Local",  "terminal": "Marathahalli Bridge",  "scheduled_arrival_time": "06:30",  "frequency_min": "25"}
```

Timestamp

8/2/2026, 12:27:27

Timestamp type: CREATE_TIME

Key Serde

String

Size: 4 Bytes

Value Serde

String

Size: 289 Bytes

+	3	18	8/2/2026, 12:27:27	R037	{ "bus_id": "BUS-0068", "route_id": "R037", "lat": 1...
+	0	11	8/2/2026, 12:27:27	R004	{ "bus_id": "BUS-0283", "route_id": "R004", "lat": 1...
+	1	11	8/2/2026, 12:27:27	R010	{ "bus_id": "BUS-0098", "route_id": "R010", "lat": 1...
+	2	11	8/2/2026, 12:27:27	R033	{ "bus_id": "BUS-0466", "route_id": "R033", "lat": 1...

We also noticed the **enrich data** in **Consumers**:

UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Consumers

Q Search by Consumer Group ID

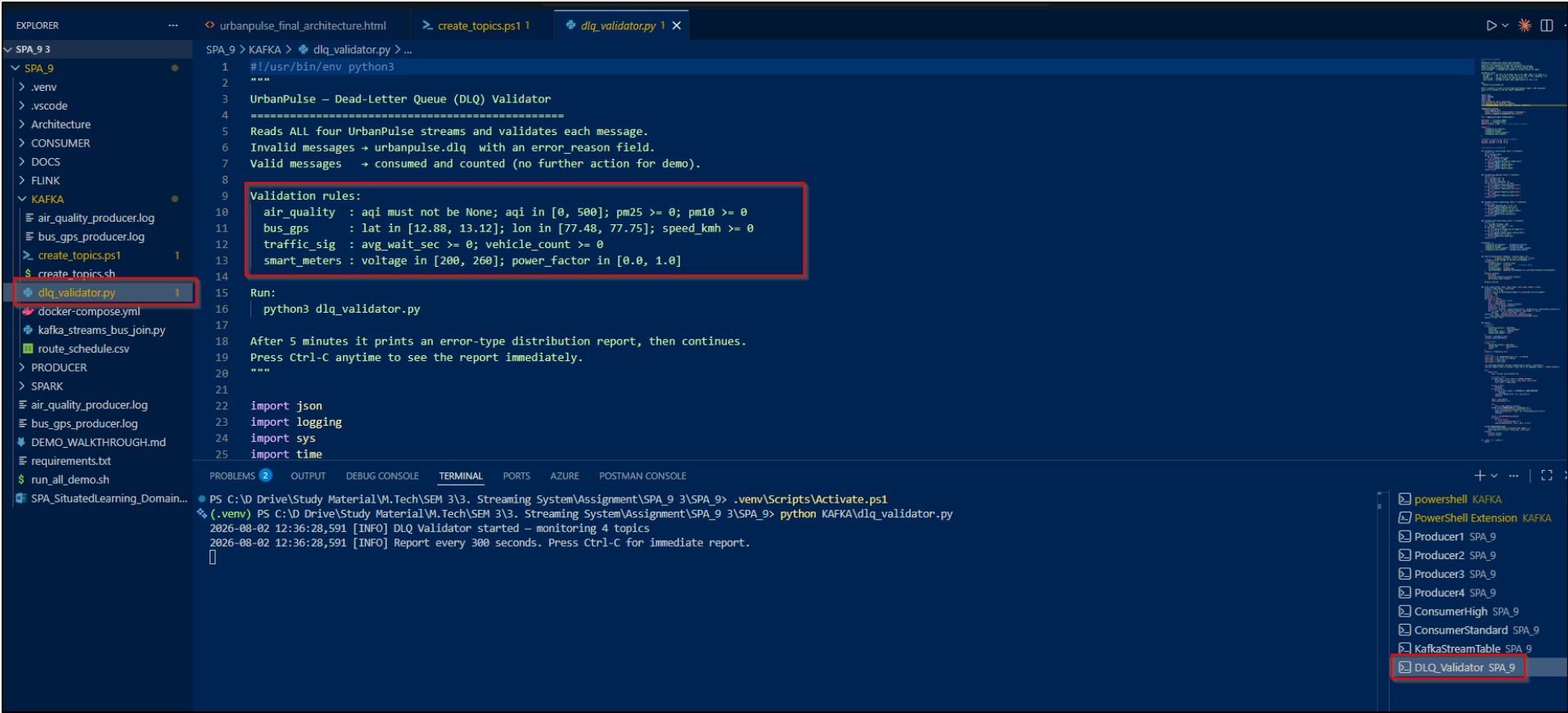
Group ID	Num Of Members	Num Of Topics	Consumer Lag	Coordinator	State
HIGH_PRIORITY_GROUP	1	1	380	3	STABLE
STANDARD_PRIORITY_GROUP	1	1	396526	1	STABLE
bus-gps-enrichment-job	1	1	2400	1	STABLE

Implement a dead-letter queue (DLQ) pattern:

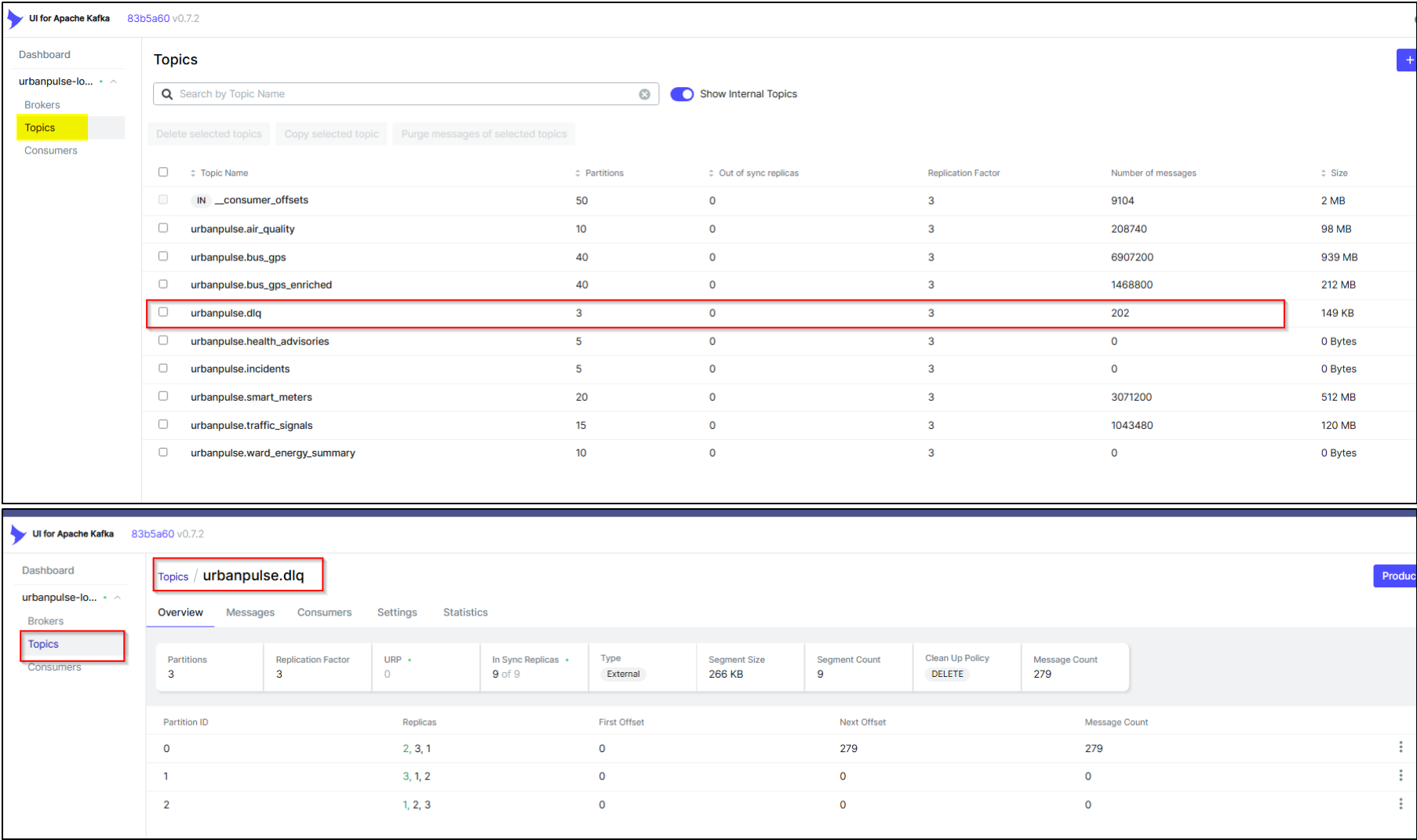
- **dlq_validator.py** — Consumes all four UrbanPulse data streams (air quality, bus GPS, traffic signals, smart meters), applying stream-specific validation rules (range checks, null checks, physical bounds) to each message.
- Invalid messages are routed to urbanpulse.dlq with the failure reason(s) attached, while valid messages are counted; a periodic error-distribution report (every 5 min, or on Ctrl-C) summarizes DLQ rates and error types per stream.

Step 7: Run the python script for dlq_validator.py

Command: python KAFKA/ dlq_validator.py



We can see the dlq data in Topics:



UI for Apache Kafka 83b5a60 v0.7.2

Dashboard

urbanpulse-lo... ^

Brokers

Topics

Consumers

Topics / urbanpulse.dlq

Overview Messages Consumers Settings Statistics

Seek Type: Offset Offset Partitions: All items are selected. Key Serde: String Value Serde: String Clear all Submit

Search + Add Filters

Offset Partition Timestamp Key Preview Value Preview

0 0 8/2/2026, 12:36:31 urbanpulse.air_quality {"original_topic": "urbanpulse.air_quality", "error_r...

Key Value Headers

```
{
  "original_topic": "urbanpulse.air_quality",
  "error_reason": "null_aqi",
  "all_errors": [
    "null_aqi"
  ],
  "original_data": {
    "sensor_id": "AQ-027",
    "zone": "East",
    "pm25": 38.7,
    "pm10": 143.2,
    "no2": 86.8,
    "aqi": null,
    "timestamp": "2026-08-02T07:06:31.923+00:00"
  },
  "dlq_timestamp": "2026-08-02T07:06:31.943+00:00"
}
```

Timestamp 8/2/2026, 12:36:31
Timestamp type: CREATE_TIME

Key Serde String
Size: 22 Bytes

Value Serde String
Size: 309 Bytes

1 0 8/2/2026, 12:36:31 urbanpulse.air_quality {"original_topic": "urbanpulse.air_quality", "error_r...

2 0 8/2/2026, 12:36:31 urbanpulse.air_quality {"original_topic": "urbanpulse.air_quality", "error_r...

3 0 8/2/2026, 12:36:32 urbanpulse.air_quality {"original_topic": "urbanpulse.air_quality", "error_r...

4 0 8/2/2026, 12:36:32 urbanpulse.air_quality {"original_topic": "urbanpulse.air_quality", "error_r...

5 0 8/2/2026, 12:36:33 urbanpulse.air_quality {"original_topic": "urbanpulse.air_quality", "error_r...

We also noticed the **dlq** data in **Consumers**:

localhost:8080/ui/clusters/urbanpulse-local/consumer-groups

UI for Apache Kafka 83b5a60 v0.7.2

Dashboard

urbanpulse-lo... ^

Brokers

Topics

Consumers

Search by Consumer Group ID

Group ID	Num Of Members	Num Of Topics	Consumer Lag	Coordinator	State
HIGH_PRIORITY_GROUP	1	1	1017	3	STABLE
STANDARD_PRIORITY_GROUP	1	1	1078444	1	STABLE
bus-gps-enrichment-job	1	1	4800	1	STABLE
dlq-validator	1	4	17297	1	STABLE

We can see the **DLQ** report in the **terminal**:

EXPLORER

SPA_9_3

SPA_9

.venv

.vscode

Architecture

CONSUMER

DOCS

FLINK

KAFKA

air_quality_producer.log

bus_gps_producer.log

create_topics.ps1

create_topics.sh

dlq_validator.py

docker-compose.yml

kafka_streams_bus_join.py

route_schedule.csv

PRODUCER

SPARK

air_quality_producer.log

bus_gps_producer.log

DEMO_WALKTHROUGH.md

requirements.txt

run_all_demo.sh

SPA_SituatedLearning_Domain...

urbanpulse_final_architecture.html

create_topics.ps1 1

dlq_validator.py 1

```
1 #!/usr/bin/env python3
2 """
3 UrbanPulse - Dead-Letter Queue (DLQ) Validator
4 =====
5 Reads ALL four UrbanPulse streams and validates each message.
6 Invalid messages -> urbanpulse.dlq with an error_reason field.
7 Valid messages -> consumed and counted (no further action for demo).
8
9 Validation rules:
10 air_quality : aqi must not be None; aqi in [0, 500]; pm25 >= 0; pm10 >= 0
11 bus_gps : lat in [12.88, 13.12]; lon in [77.48, 77.75]; speed_kmh >= 0
12 traffic_sig : avg_wait_sec >= 0; vehicle_count >= 0
13 smart_meters : voltage in [200, 260]; power_factor in [0.0, 1.0]
14
15 Run:
16 python3 dlq_validator.py
17
18 After 5 minutes it prints an error-type distribution report, then continues.
19 Press Ctrl-C anytime to see the report immediately.
20 """
21
22 import json
23 import logging
24 import sys
25 import time
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE POSTMAN CONSOLE

PS C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9 3\SPA_9> .venv\Scripts\Activate.ps1

(.venv) PS C:\D Drive\Study Material\VM.Tech\SEM 3\3. Streaming System\Assignment\SPA_9 3\SPA_9> python KAFKA\dlq_validator.py

2026-08-02 12:36:28,591 [INFO] DLQ Validator started - monitoring 4 topics

2026-08-02 12:36:28,591 [INFO] Report every 300 seconds. Press Ctrl-C for immediate report.

```
=====
DLQ Error Distribution Report - 5.0 min window
=====
[air_quality] total=17820 valid=16927 dlq=893
|- null_aqi : 893

[bus_gps] total=712800 valid=712800 dlq=0

[traffic_signals] total=112480 valid=112480 dlq=0

[smart_meters] total=326700 valid=326700 dlq=0

TOTAL valid=1168907 dlq=893 dlq_rate=0.1%
=====
```

powerShell KAFKA

PowerShell Extension KAFKA

Producer1 SPA_9

Producer2 SPA_9

Producer3 SPA_9

Producer4 SPA_9

ConsumerHigh SPA_9

ConsumerStandard SPA_9

KafkaStreamTable SPA_9

DLQ_Validator SPA_9